



SCI_BUILD_A2_W03 · BUILD Science teacher note

Staff-facing; review only. Verbatim SoW objective: Test rocks for hardness and whether they let water through.

Source: BUILD scheme of work, Autumn 2 week 3. Pupil goal: I am learning to test a rock in two ways.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can show a safe mark test., I can watch where drops go., I can tell what I saw.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	How can we sort two rocks?	Check a property you can see or feel.	BUILD scheme of work, Autumn 2 week 2
2-4 weeks	What can we do with a clean tub?	Use it again if safe.	BUILD scheme of work, Autumn 2 week 1
last term	Name one body part that helps protect organs.	A skeleton or bones.	BUILD scheme of work, Autumn 1 week 1

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Is a wet top proof that water went right through? Answer: No. Look beneath the rock or in the tray to check whether any water passed through. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
hardness	Tier 3	how hard a thing is to mark	The tool marks this rock.
absorb	Tier 3	take in a liquid	Water soaks in.
water	Tier 2	a liquid used in our test	I add three drops.
test	Tier 2	a check to find out	I test two rocks.

Faded example, tiered task and stretch

I do full: "Adult uses one blunt tool. A gets a mark. Three drops stay on top." We do part-done: "Mark B. Add three drops. The drops ." Removed step: State the observed result at rock B, rather than guess it. Independent: no filled step. Next fade: Pupil picks the test step and result from the unlabelled kit.

We do F: Watch one drop. Point to the top or tray. M: Try the safe test. Say what happened. H: Why must each rock get the same drops? Independent F: Point to a mark and a wet rock. M: Show both tests. Tell what you saw. H: Why use the same tool and drops? Go deeper: Could a hard rock also let water through? Use each test result. Named harder stretch — Run-off check: A drop slides off. How will you fix the test?



The kit route is hands-on or sensory: safe rocks, adult-held blunt tool, dropper, tray and towel. Two on-screen interactions are card sort and What must you check to say water passed through?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Will drops pass through? Kit: Two safe rocks, Adult-held blunt tool, Dropper and tray. Steps: Guess first; Adult tests a mark on each rock; Add three drops to A. Check the edge and below; Do the same on B. Record the route seen. Results headings: Rock, Mark?, Water route seen. Check against the school risk assessment and CLEAPSS guidance. Adult handles tool, inspects edges and controls water; no dust or tasting.

Hinge and decision

Question: The tray under a new rock C is wet before the water test. What should the class do first?

Option	Correct?	Diagnosis
Call C porous from the wet tray.	no	Uses water present before the test as a result.
Call C hard from the wet tray.	no	Confuses a water clue with hardness.
Skip the test and guess from colour.	no	Uses appearance instead of the planned check.
Dry the tray, then repeat the test.	yes	Removes the old water before observing a new result.

Rule: About 3 in 4 or more right → move on; otherwise re-model Adult uses one blunt tool. A gets a mark. Three drops stay on top. Ask the parallel hinge. No scores or answers saved. Parallel hinge: A drop from new rock D runs around its edge into a dry tray. Does that prove water passed through D? No; check the route and repeat safely.

Misconceptions:

Misconception	Spot	Correct
A wet surface proves water went through a rock.	Points to a wet top as the result.	Check the underside and tray; state only what was observed.
The scratch test and water test measure the same thing.	Uses scratch result to answer a water question.	Compare two separate result columns, mark and water through.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ? At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ? R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Give a three-step visual card for test rock hardness and water; remove its last cue after a correct try.
Communication & Interaction	explicit instruction	Point to the real kit before a short spoken, signed or AAC response; wait for a turn.



Broad area	EEF five-a-day strategy	Specific support
SEMH	cognitive/metacognitive strategies	Offer a private first-then cue and planned pause before the hinge; re-enter with one step.
Sensory/Physical	technology	Use a stable tray, large cards or an accessible input; no forced touch.
EAL (language, not SEN)	flexible grouping	Rehearse a full M or H idea in any language before writing; keep the science challenge.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the object or model. → Ask: What will you check next? → Model one step and give the next turn back. → Finish one example together, then let the pupil try. No physical prompt. Stage scripts: I do “Watch my first step. What do you see?”; We do “What step is missing? Show it with the kit.”; Independent “What can you do first on your own?”. At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Point to the tray below the rock. → The tray that catches drops beneath it; What is a test? → A check to find out.
- M: Does a wet top prove water went through? → No. Check beneath the rock and tray; What does absorb mean? → Take in a liquid.
- H: Why use the same tool and drops? → Then the rock is the planned change in the check; What does a drawing leave out? → It cannot show whether drops truly passed through a sample.

Regulation: Agree a quiet pause and supervised place. Give one short cue. Re-enter at the last completed step. No evidence while distressed.

Lesson record: BUILD_SCI_A2_W03 via staff device or projector only. Gatsby 4: A lab technician compares rock samples with the same checks.

Reading gate: BUILD below age 8; rough ARI grade 0.61 on 648 pupil-visible words (adult guidance and navigation excluded); heuristic age guide 6.61 years, not a formal reading-age test. This is a rough screen, not an individual assessment.

Apparatus staff to confirm: The USGS source describes water moving through connected spaces in rock. It does not validate our open classroom tray. Inspect whether drops travel through the actual sample or around its edge; record the observed route without claiming permeability from tray drops alone.

Pupil-facing risk: Tools or rough rocks may hurt skin; spills may be slippery.

Pupil-facing safety: Let staff use the tool. Keep drops in the tray. Do not taste; wash hands after.

SLT planning and evaluation

Lesson code: SCI-BUILD-A2-W03

SoW reference: BUILD scheme of work, Autumn 2 week 3

Exact SoW outcome: Test rocks for hardness and whether they let water through.



Evidence (scheme of work): Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Entry Level Science 8939 (E1→E3 intro); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-BUILD-A2-W03. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Point to a safe rock. Staff: Offer a safe kit choice and short first-then cue.

Starter (3 min): Think back. Say, sign or point. Staff: Ask three low-stakes prior questions; answers in note.

I Do (5 min): Watch two safe tests. Staff: Pre-teach words; show the full model; ask the show-me check.

We Do (7 min): Test two safe rocks. Staff: Hand over the missing step; offer F or H route. Reset once.

We Do 2 (5 min): All of us show one choice. Staff: Take all pupil responses at once; use decision rule and parallel hinge.

Independent (8 min): Show each test. Record what you saw. Staff: Choose one adult for supported pupils, one for group; swap across the week.

Review (4 min): Check one result. Hear a next step. Staff: Give VF and one NS(+); invite one self-edit. Reset once.

Exit (3 min): Tell an adult what you saw. Staff: Hear feedback and next step, choose next start, then one-minute reset.

Planned adaptation — cognition learning: Give a three-step visual card for test rock hardness and water; remove its last cue after a correct try.

Planned adaptation — communication interaction: Point to the real kit before a short spoken, signed or AAC response; wait for a turn.

Planned adaptation — semh: Offer a private first-then cue and planned pause before the hinge; re-enter with one step.

Planned adaptation — sensory physical: Use a stable tray, large cards or an accessible input; no forced touch.

Planned adaptation — eal: Rehearse a full M or H idea in any language before writing; keep the science challenge.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: explicit instruction; semh: cognitive/metacognitive strategies; sensory physical: technology; eal: flexible grouping

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Show each test. Record what you saw. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W04. Keep the core evidence and any safety teaching; record the change and coverage impact.



If behind — next lesson: Autumn 2 W04: Identify everyday uses of different rocks (links D&T / Construction).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: State the observed result at rock B, rather than guess it. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "The tray under a new rock C is wet before the water test. What should the class do first?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Point to a safe rock. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Show each test. Record what you saw. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use hardness, absorb, water, test with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.